Real-Pair Representations of Complex Numbers and Quaternions:
Algebra, Arithmetic Coordinates, and Implementations in R and Python

Fred Viole

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Abstract

We study the real-pair transform $\phi(a+bi) = (a-b, a+b)$ and its blockwise extension to quaternions. These maps are scaled orthogonal changes of coordinates; complex multiplication and quaternion multiplication are transported to explicitly defined products on $\mathbb{R}^2$ and $\mathbb{R}^4$, rather than ordinary componentwise multiplication. We give exact identities for conjugation, multiplication, inversion, norms and exponentials. We also identify two specific arithmetic interpretations: $\phi(\mathbb{Z}[i])$ is an index-two parity sublattice, and the inverse map turns a factor pair (c, q) into a Fermat midpoint/half-gap complex coordinate whose square has real part cq. We describe the corresponding real-coordinate form of zeta-zero reflection without asserting any new zero-free theorem. Reproducible implementations in base R and standard-library Python provide complex and quaternion operations, worked examples, and deterministic and randomized verification. Together, the results give an exact coordinate dictionary for arithmetic, algebraic, and spectral computations.

1 Introduction and scope

Representing complex numbers by real coordinate pairs is a standard algebraic technique. This paper develops the particular transform $(a, b) \mapsto (a-b, a+b)$ as a common coordinate system for complex and quaternion arithmetic, integer factor-pair geometry, and spectral calculations. Explicit transported operations make the representation usable in real-valued computational environments, including R and Python.

The selected coordinates are especially natural for factor-pair bookkeeping: they recover the two factors in Fermat's difference-of-squares parameterization directly from midpoint and half-gap coordinates. The same representation preserves normalized Gram products and expresses complex spectral modes through real vectors without altering their zeros or analytic properties. Sections 2–4 prove the identities; Sections 3 and 5 explain the arithmetic and spectral interpretations. Code and verification procedures and the companion source archive are described in Section 6.

2 Complex arithmetic in real-pair coordinates

Define

$$\phi : \mathbb{C} \longrightarrow \mathbb{R}^2, \qquad \phi(a+bi) = \begin{pmatrix} a-b \\ a+b \end{pmatrix}, \qquad T = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}. \tag{1}$$

Since $T^{\mathsf{T}}T = 2I$ and $\det T = 2$, the inverse is

$$\phi^{-1}(x, y) = \frac{x+y}{2} + i\frac{y-x}{2}. \tag{2}$$

Thus $T/\sqrt{2}$ is orthogonal, and ϕ is an invertible real-linear mapping (a 45° rotation followed by $\sqrt{2}$ scaling).

Theorem 2.1 (Transported complex field)**.** *Equip $\mathbb{R}^2$ with ordinary vector addition, the multiplicative identity $e = (1, 1)$, and the product*

$$(x, y) \odot (u, v) = \left(\frac{xv + yu - yv + xu}{2}, \frac{xv + yu + yv - xu}{2} \right). \tag{3}$$

Then $\phi : \mathbb{C} \to (\mathbb{R}^2, +, \odot)$ is a field isomorphism. Complex conjugation and squared modulus become

$$\phi(\overline{z}) = (y, x), \qquad |z|^2 = \frac{x^2 + y^2}{2}, \quad (x, y) = \phi(z). \tag{4}$$

For $(x, y) \neq (0, 0)$,

$$(x, y)_{\odot}^{-1} = \frac{2}{x^2 + y^2}(y, x). \tag{5}$$

Proof. Write $a = (x + y)/2$ and $b = (y - x)/2$, with corresponding $c = (u + v)/2$ and $d = (v - u)/2$. Complex multiplication yields $(ac - bd) + i(ad + bc)$. Applying (1) and expanding gives (3). Since this is the transported product, associativity, distributivity, and inverses follow from those of $\mathbb{C}$. The real unit 1 maps to $(1, 1)$, and i maps to $(-1, 1)$. Replacing b by $-b$ swaps x, y. Finally, $(a - b)^2 + (a + b)^2 = 2(a^2 + b^2)$, which gives the norm and inverse formulas. □

Remark 2.2. *The expression "field isomorphism" refers to $\mathbb{R}^2$ with the* transported *product $\odot$. With ordinary componentwise multiplication, $\mathbb{R} \times \mathbb{R}$ is not a field, so the unqualified statement that ϕ preserves multiplication in standard $\mathbb{R}^2$ would be incorrect.*

For $z = a + bi$ and $\phi(z) = (x, y)$, the transported exponential is

$$\phi(e^z) = e^a(\cos b - \sin b,\ \cos b + \sin b). \tag{6}$$

Furthermore, if $m = \alpha + i\beta$ and $v = \phi(z)$, then

$$\phi(mz) = \begin{pmatrix} \alpha & -\beta \\ \beta & \alpha \end{pmatrix} v, \qquad \frac{\|\phi(mz)\|_2^2}{2} = |m|^2 \frac{\|v\|_2^2}{2}. \tag{7}$$

These identities permit complex spectral multipliers to be applied to a pair of real coordinates while preserving their normalized energy.

3 Arithmetic meaning of the chosen transform

The map is not a new field, but the choice of coordinates has useful arithmetic interpretations.

Proposition 3.1 (Gaussian-integer parity lattice)**.** *The image of the Gaussian integers is*

$$\phi(\mathbb{Z}[i]) = \{(x, y) \in \mathbb{Z}^2 : x \equiv y \pmod 2\}. \tag{8}$$

It is an additive sublattice of index two. The Gaussian norm is

$$N(a + bi) = a^2 + b^2 = \frac{x^2 + y^2}{2}, \tag{9}$$

which is multiplicative under $\odot$.

Proof. If $a, b \in \mathbb{Z}$, then $x = a - b$ and $y = a + b$ have the same parity. Conversely, if $x \equiv y \pmod 2$, then $(x + y)/2$ and $(y - x)/2$ are integers. Exactly two parity classes of $\mathbb{Z}^2$ exist under the homomorphism $(x, y) \mapsto x - y \pmod 2$, giving index two. The norm identity is (4); its multiplicativity follows from $|zw|^2 = |z|^2|w|^2$. □

Proposition 3.2 (Fermat factor-pair coordinates)**.** *For real numbers c, q set*

$$z(c, q) = \frac{c + q}{2} + i\frac{q - c}{2}. \tag{10}$$

Then the factor pair is recovered exactly*:*

$$\phi(z(c, q)) = (c, q), \qquad z(c, q)^2 = cq + i\frac{q^2 - c^2}{2}. \tag{11}$$

In particular, if $n = cq$, the real part of $z(c, q)^2$ is n. If c, q are integers of the same parity, then $z(c, q) \in \mathbb{Z}[i]$.

Proof. The first identity is the substitution of (10) into (1). The real part of the square is $((c + q)/2)^2 - ((q - c)/2)^2 = cq$; its imaginary part is $2((c + q)/2)((q - c)/2) = (q^2 - c^2)/2$. The parity assertion follows from (8). □

For integer a, b, this is precisely the difference-of-squares identity $(a - b)(a + b) = a^2 - b^2$, expressed with the two factors as the output coordinates. This is a reason to prefer T over the standard (a, b) coordinates for *factor-pair bookkeeping*; it is not a faster factorization algorithm or a new prime-distribution theorem. The Gaussian-integer context is classical; see [2, 3].

4 Quaternion arithmetic in real quadruples

Use the quaternion convention $i^2 = j^2 = k^2 = ijk = -1$ and identify

$$q = a + bi + cj + dk = z_1 + z_2 j, \qquad z_1 = a + bi, \quad z_2 = c + di. \tag{12}$$

Define the blockwise transform

$$\psi(q) = (a - b, a + b, c - d, c + d) = (\phi(z_1), \phi(z_2)) \in \mathbb{R}^4. \tag{13}$$

It is an invertible real-linear map. It is not an algebra isomorphism for ordinary componentwise multiplication: quaternion multiplication must be transported, and it is noncommutative.

Theorem 4.1 (Transported quaternion operations)**.** *With $q = z_1 + z_2 j$ and $r = w_1 + w_2 j$, quaternion multiplication is*

$$qr = (z_1 w_1 - z_2 \overline{w_2}) + (z_1 w_2 + z_2 \overline{w_1})j. \tag{14}$$

Applying ϕ to both complex components defines the corresponding product $\star$ on $\mathbb{R}^4$. If $p = \psi(q) = (x_1, y_1, x_2, y_2)$, then

$$\boxed{\psi(\overline{q}) = (y_1, x_1, -x_2, -y_2)} \tag{15}$$

and

$$\|q\|_{\mathbb{H}}^2 = a^2 + b^2 + c^2 + d^2 = \frac{x_1^2 + y_1^2 + x_2^2 + y_2^2}{2}. \tag{16}$$

For $q \neq 0$ the transported inverse is

$$\boxed{p_\star^{-1} = \frac{2}{x_1^2 + y_1^2 + x_2^2 + y_2^2}(y_1, x_1, -x_2, -y_2).} \tag{17}$$

Proof. The relation $jz = \overline{z}j$ for $z \in \mathbb{C}$ and $j^2 = -1$ gives (14) by expansion; this is the standard complex-pair form of quaternion multiplication [1]. Quaternion conjugation sends $z_1 + z_2 j$ to $\overline{z_1} - z_2 j$. Complex conjugation swaps the entries of $\phi(z_1)$, while $-z_2$ negates *both* entries of $\phi(z_2)$, proving (15). The norm follows by two applications of (4). Finally $q^{-1} = \overline{q}/\|q\|_{\mathbb{H}}^2$ for nonzero quaternions, which gives (17). □

Conjugation, inverse, and right division. The first real pair of a quaternion is complex-conjugated by swapping its two entries; the second real pair changes sign, as follows from $\overline{z_1 + z_2 j} = \overline{z_1} - z_2 j$. For example, $\psi(j) = (0, 0, 1, 1)$ and $\psi(j^{-1}) = (0, 0, -1, -1)$, since $j^2 = -1$. Quaternion multiplication is noncommutative, so right division is defined by $p/q = p \star q_\star^{-1}$.

For $q = a + \mathbf{v}$, with $\mathbf{v} = bi + cj + dk$ and $r = \sqrt{b^2 + c^2 + d^2}$, the quaternion exponential is

$$\exp(q) = e^a \left(\cos r + \frac{\sin r}{r} \mathbf{v} \right), \qquad \left. \frac{\sin r}{r} \right|_{r=0} = 1. \tag{18}$$

The companion code uses the fourth-order Taylor polynomial near zero instead of setting the vector component to zero, thereby preserving small nonzero imaginary components.

5 Real coordinates for complex spectral parameters

The same transform can encode a complex spectral parameter without changing its analytic meaning. If $\rho = \beta + i\gamma$ and $(u, v) = \phi(\rho)$, then

$$u = \beta - \gamma, \qquad v = \beta + \gamma, \qquad \Re\rho = \tfrac{1}{2} \iff u + v = 1. \tag{19}$$

The reflection $\rho \mapsto 1 - \overline{\rho}$ becomes

$$(u, v) \longmapsto (1 - v, 1 - u). \tag{20}$$

These are exact coordinate statements; they do not place any zeros on the critical line.

For an illustrative homogeneous factor $m = q^{2\rho - 1}$ with $q > 0$, Equation (7) gives the real-pair squared-energy multiplier

$$|m|^2 = q^{4\beta - 2}. \tag{21}$$

At $\beta = 1/2$ this multiplier is one; at $\beta > 1/2$ it is greater than one for $q > 1$. Equation (21) concerns an individual complex multiplier; turning it into a bound on an actual prime- or Möbius-weighted sum requires additional arithmetic estimates.

Lean formalizations of the real-pair interface, its Gram identity, and the factor-pair correspondence are available in the RH_Lean repository, notably realification module and Fermat-coordinate module. That project also includes a conditional square-endpoint reformulation of the von Koch prime-counting criterion. The present note does *not* prove the Riemann hypothesis or an improved prime-counting error bound; those are separate questions.

6 Implementation and verification

The supplied files `Real_Pair_Representations.R` and `Real_Pair_Representations.py` implement complex real-pair and quaternion real-quadruple arithmetic using base R and the Python standard library, respectively. Conversions to and from each language's native complex type are restricted to `phi` and `phi_inv`; the remaining arithmetic functions use real-valued vectors or tuples.

This facilitates use with numerical code expecting real vectors, although no speed advantage over native complex operations is claimed.

Two numerical details matter. First, norm-squared expressions are scaled by the largest absolute entry before inversion, reducing avoidable intermediate overflow/underflow in squared terms (without guaranteeing accuracy at every floating-point extreme). Second, the quaternion exponential evaluates $\sin r/r$ through its Taylor series near zero. We use exact zero checks for algebraic inverses rather than silently rejecting valid small nonzero quaternions by an arbitrary absolute threshold.

Table 1: Properties checked by the companion R and Python test suites.

Check	Assertion
Complex transport	Multiplication, division and exponential agree with native R and Python complex arithmetic.
Quaternion basis	$ij = k$, $ji = -k$, $j^2 = -1$.
Quaternion inverses	$j^{-1} = -j$, $k^{-1} = -k$; left and right inverse identities.
Right division	$k/i = -j$.
Quaternion exponential	$e^0 = 1$, $e^{\pi j} = -1$, and a small nonzero j component.
Randomized comparisons	1000 seeded cases in R and 10,000 seeded cases in Python against the independent quaternion coefficient formula; norm multiplicativity.

The R suite uses `stopifnot`, and the Python suite uses `unittest`. The test sources include deterministic quaternion basis cases, both-sided inverses, Gaussian norms, factor-pair identities, complex exponentials, and seeded randomized comparisons to an independent quaternion coefficient formula. The Python tests require no third-party packages. All implementations and verification scripts are freely available in the Number-Theory companion code directory.

Language	Implementation	Test suite
R	Source (.R)	Tests (.R)
Python	Source (.py)	Tests (.py)

The repository also provides worked Python examples, independent numerical checks, and a README with execution instructions.

7 Conclusion

The transform $\phi(a + bi) = (a - b, a + b)$ is a classical linear change of coordinates equipped with explicitly transported operations. Its concrete value here lies in preserving complex and quaternion algebra using real vectors, representing Gaussian integers as a parity lattice, and recovering Fermat factor pairs directly from the midpoint/half-gap complex coordinate. Base-R and standard-library Python implementations make the operations computationally reproducible, with explicit treatment of quaternion inversion and the exponential near zero. Both companion test suites exercise the transported algebra through deterministic examples and randomized identities. These coordinate relations can support further arithmetic and spectral analyses, while a factorization algorithm or a new zero-free theorem would require additional mathematical arguments.

References

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[4] H. von Koch, "Sur la distribution des nombres premiers," *Acta Mathematica* 24 (1901), 159–182.

[5] OVVO Financial, *RH_Lean: Square-block prime and Möbius formalizations*, 2026, https://github.com/OVVO-Financial/RH_Lean.